

Roll No.....

ODD MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: B.Tech

Semester: III

Name of the Course: Thermal Engineering

Course Code: TME 308

Time: 90 Minute

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing any of the sub questions.
- (ii) Each question carries 10 marks.

Q.1	(10 marks)																							
(a)	Distinguish between: (i) Intensive and extensive properties, (ii) Open and closed system, (iii) Thermal and mechanical equilibrium, and (iv) Heat and internal energy.	CO1																						
OR																								
(b)	With the help of p-v diagram, derive the work transfer for adiabatic process.	CO1																						
Q.2	(10 marks)																							
(a)	To a closed system 150 kJ of work is supplied. If the initial volume is 0.6 m^3 and pressure of the system changes as $p = 8 - 4V$, where p is in bar and V is in m^3 , determine the final volume and pressure of the system.	CO1																						
OR																								
(b)	A vessel of volume 0.04 m^3 contains a mixture of saturated water and saturated steam at a temperature of 250°C . The mass of the liquid present is 9 kg. Find the total mass, the specific volume, the enthalpy, the entropy, and the internal energy.	CO1																						
<table><tr><td>Temperature $^\circ\text{C}$</td><td>Pressure MPa</td><td colspan="2">Specific volume m^3/kg</td><td colspan="2">Specific enthalpy kJ/kg</td><td colspan="2">Specific entropy kJ/kg K</td></tr><tr><td rowspan="2">250</td><td rowspan="2">3.973</td><td>V_f</td><td>V_g</td><td>h_f</td><td>h_{fg}</td><td>S_f</td><td>S_{fg}</td></tr><tr><td>0.0012512</td><td>0.05013</td><td>1085.36</td><td>1716.2</td><td>2.7927</td><td>3.2802</td></tr></table>			Temperature $^\circ\text{C}$	Pressure MPa	Specific volume m^3/kg		Specific enthalpy kJ/kg		Specific entropy kJ/kg K		250	3.973	V_f	V_g	h_f	h_{fg}	S_f	S_{fg}	0.0012512	0.05013	1085.36	1716.2	2.7927	3.2802
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Q.3	(10 marks)																							
(a)	What is a pure substance? Draw and explain a p-T (pressure-temperature) diagram for a pure substance.	CO1																						
OR																								
(b)	Derive the steady flow energy equation for a single stream entering and a single stream leaving a control volume and explain the various terms in it.	CO2																						
Q.4	(10 marks)																							
(a)	The working fluid, in a steady flow process flows at a rate of 220 kg/min . The fluid rejects 100 kJ/s passing through the system. The conditions of the fluid at inlet and outlet are given as: $V_1 = 320 \text{ m/s}$, $p_1 = 6 \text{ bar}$, $u_1 = 2000 \text{ kJ/kg}$, $v_1 = 0.36 \text{ m}^3/\text{kg}$ and $V_2 = 140 \text{ m/s}$, $p_2 = 1.2 \text{ bar}$, $u_2 = 1400 \text{ kJ/kg}$, $v_2 = 1.3 \text{ m}^3/\text{kg}$. Determine the power capacity of the system in MW.	CO2																						
OR																								
(b)	A house requires $2 \times 10^5 \text{ kJ/h}$ for heating in winter. Heat pump is used to absorb heat from cold air outside in winter and send heat to the house. Work required to operate the heat pump is $3 \times 10^4 \text{ kJ/h}$. Determine (i) Heat abstracted from outside and (ii) Co-efficient of performance.	CO2																						

Q.5	(10 marks)	
(a)	A turbine, operating under steady-flow conditions, receives 4500 kg of steam per hour. The steam enters the turbine at a velocity of 2800 m/min, an elevation of 5.5 m and a specific enthalpy of 2800 kJ/kg. It leaves the turbine at a velocity of 5600 m/min, an elevation of 1.5 m and a specific enthalpy of 2300 kJ/kg. Heat losses from the turbine to the surroundings are 16000 kJ/h. Determine the power output of the turbine.	CO2
OR		
(b)	A Carnot heat engine draws heat from a reservoir at temperature T_1 and rejects heat to another reservoir at temperature T_3 . The Carnot forward cycle engine drives a refrigerator which absorbs heat from reservoir at temperature T_2 and rejects heat to a reservoir at temperature T_3 . If the high temperature $T_1 = 600$ K and low temperature $T_2 = 300$ K, determine (i) The temperature T_3 such that heat supplied to engine Q_1 is equal to the heat absorbed by refrigerator Q_2 and (ii) The efficiency of Carnot engine and COP of refrigerator.	CO2